

Fully Automated High-Throughput Battery Development

Example Workflows for Battery Research using the Chemspeed FLEXSHUTTLE

The global demand for advanced energy storage solutions is rapidly increasing driven by the proliferation of electric vehicles, portable electronics and energy storage technologies. Battery research and development plays a huge role in meeting these demands. However, manual battery development processes are time consuming, labour-intensive and prone to inconsistencies. This necessitates the adoption of high-throughput, automated approaches to accelerate and optimise battery research processes.

This application note provides an overview of the Chemspeed's FLEXSHUTTLE platform and its integrated workflow management ARKSUITE software for high-throughput automated battery development processes.

Keywords: Battery, automated, workflows, high-throughput, Chemspeed FLEXSHUTTLE

Introduction

The surge in demand for efficient and sustainable energy storage devices has driven the need for advanced batteries with enhanced performance, longevity and safety. Traditional methods are often slow due to extensive manual experimentation, testing and not to mention, prone to errors.

This article highlights Chemspeed's fully automated solution for battery research. Example workflows on the FLEXSHUTTLE platform shows how each step significantly accelerates battery development lifecycle.

Application Showcase

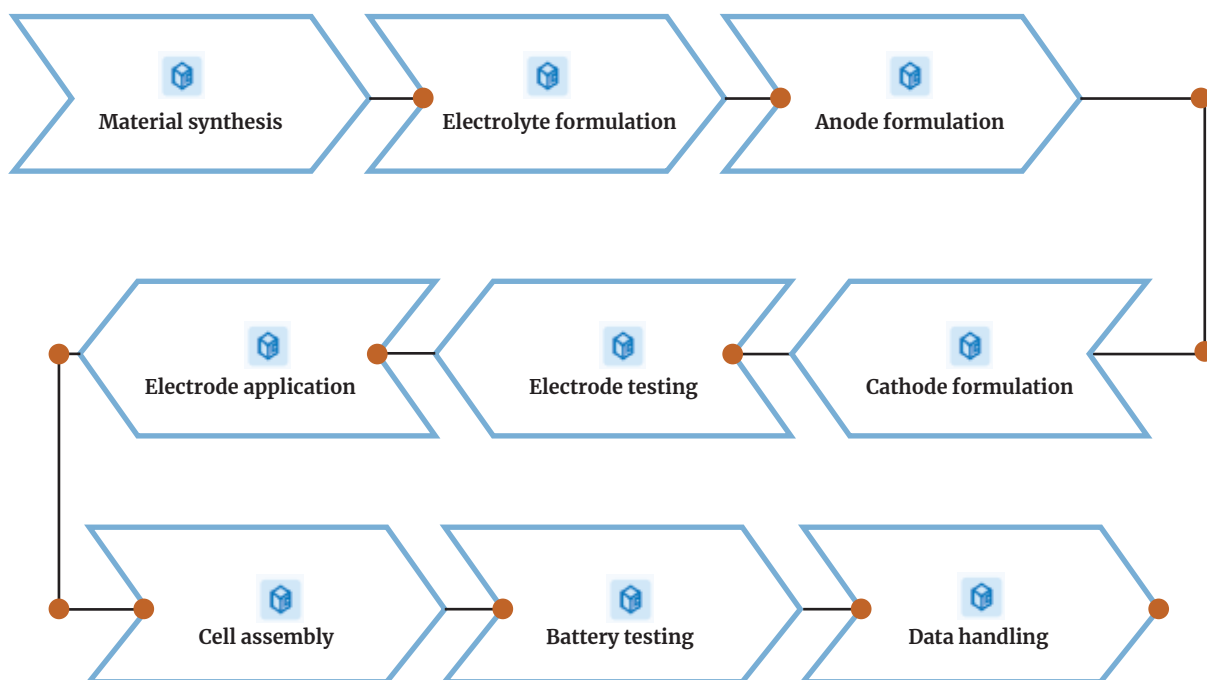
Chemspeed's solution

The modular design enables significant efficiency increase combined with high quality and maximum ease of use.

- › Design your automated laboratory
- › Navigate on the system with shuttles
- › Model, execute and control multiple workflows
- › Ensure conditioned atmosphere as required by means of tight enclosure
- › Dispense gravimetrically liquids, pastes and solids from small to large amounts in high accuracy
- › Formulate battery slurries (anode and cathode)
- › Measure and adjust rheology
- › Apply battery slurry via roll-to-roll coating followed by heat curing and calendering
- › Punch electrodes in different formats and assemble them in coin or pouch cells
- › Characterize conductivity, rechargeability and other battery properties

Example Workflows

The workflow below showcases an example workflow indicating generic steps for battery development.



Application Showcase

Example Workflow with detailed steps for coin batteries-



Application Showcase

Action	Tools and Function
Load vessel and read barcode	Multigripper Robotic Tool for Vials/Plates/ISYNTH Drawers - For transfer of vials, centrifuge tubes etc
Open vessel	Screw capping / Decapping Robotic Tool / Screw Capper - For opening, closing and discarding vials and caps
Dispense liquid (viscous)	Overhead gravimetric dispensing unit for highly viscous liquids / GDU-HV - For direct gravimetric dispensing into a target as a formulation vessel
Mix	FLEX MIX (Dual Asymmetric Centrifugation) - Fully integrated mixing system for dispersing / mixing a variety of materials: With a large range of viscosities in any combination - e.g. colloids, fluids, powders, pastes, creams, grease, resins, inks and paints mixtures
Particle size	FLEX PARTICLE SIZE (dynamic light scattering) - Automatic system for particle size measurement
Rheology	FLEX RHEOLOGY - Fully integrated automatic system for rheology measurement of liquid samples
Density	FLEX DENSITY - Automatic system for specific gravity / density measurement of liquid samples
Draw down	FLEX DRAW DOWN & HEAT CURING - Automatic roll-to-roll draw down system for metallic foil, including in-line heat curing
Heat cure	FLEX DRAW DOWN & HEAT CURING - Automatic roll-to-roll draw down system for metallic foil, including in-line heat curing
Thickness	FLEX ELECTRODE CHARACTERIZATION - Fully integrated automatic system for coating characterization: surface loading, thickness, resistance and resistivity measurement device
Surface load	FLEX ELECTRODE CHARACTERIZATION - Fully integrated automatic system for coating characterization: surface loading, thickness, resistance and resistivity measurement device
Resistivity	FLEX ELECTRODE CHARACTERIZATION - Fully integrated automatic system for coating characterization: surface loading, thickness, resistance and resistivity measurement device
Calendering	FLEX CALENDERING - Automatic system for high precision calendering of electrodes
Punch electrode	FLEX PUNCHING Fully automated system for electrode punching.
Assemble coin cell	FLEX COIN CELL ASSEMBLY Fully automated system for coin cell assembly
Store cell	FLEX COIN CELL ASSEMBLY Fully automated system for coin cell assembly
Close and store vessel	FLEX VESSEL STORAGE Automatic storage and handling system for formulation vessels
Export data	DASHBOARD Setup for data and workflow management, including real-time visualization

Powerful advantages

- › Increase throughput
- › Enhance reproducibility and quality
- › Reduce raw material consumption and waste
- › Enable digitalization
- › Release brain power for other tasks
- › Boost innovation

ARKSUITE

Workflow management software for automated systems.

ARKSUITE is an integrating software platform that serves as a central control centre for orchestrating multiple instances (incl. 3rd-party devices) – global experiment design, experiment execution, scheduling, inventory management, data management, digital twin.

Application Showcase

Integrating additional services such as LIMS, web services, and other auxiliary tools further enhance the overall process. The inclusion of 3rd -party devices e.g. for analysis / testing completes the workflow provides full end-to-end control and comprehensive data capture.

- › **Flexibility:** Highly flexible design supports essentially any interface using IoT and Internet 4.0 concepts.
- › **Data capture:** Comprehensive experimental data capture, including universal execution time stamps.
- › **Collaboration:** Central database services support ease of linking multiple systems, labs and even sites for uninhibited data share and collaboration (physical and / or cloud-based server solutions).
- › **Exceptional control:** Radical Digital Twin provides real-time top-level viewing and control of integrated systems and/or labs.



Conclusion

By automating key processes like slurry formation, electrode production to cell assembly and characterization, Chemspeed's FLEXSHUTTLE significantly increases throughput, reproducibility, quality and reduces material waste. The integrated workflow management ARKSUITE software allows seamless control and data capture enabling researchers to focus on innovation rather than manual tasks. This comprehensive solution paves the way for accelerated battery development.



FLEXSHUTTLE

Step-changing solution for battery development and quality control workflows.